DOSAGE FORMS OF DRUGS

1- Enteral (GIT).
2- Parenteral (Injection).
3- Inhalation (Respiratory tract).
4- Topical (External).

I. ENTERAL DOSAGE FORMS

Liquid | Solid | Rectal

LIQUID ORAL FORM

I. Solutions

Aqueous	Alcoholic	
Syrup	**Tincture**	**Elixir**
Sweetened aqueous solution (Sugar + H_2o + drug) Cough syrup	**Hydroalcoholic solution** (10-20% alcohol in H20) (Alcohol + H_2o + drug) Tincture belladonna	**Sweetened** hydro alcoholic solution. (Alcohol + H_2o + sugar + drug) Antihistaminic elixir
		

II. Suspension

Insoluble solid in water + Suspending agent: gum
Must be shacked before use



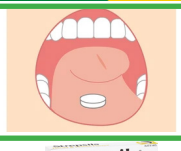
Drug soluble in

- **Water** → +Sugar → **Syrup**
- **Alcohol** → water+ → **Tincture**
- **Alcohol** → +Sugar + Water → **Elixir**

Drug insoluble in water: Suspension

SOLID ORAL FORM

A. TABLETS

1. Ordinary	A smal disc of drug mixed with **inert binder** as starch compressed together. 
2. Coated	A sugar coat is added to mask bad taste. 
3. Enteric coated	The coat is disintegrated in the intestine to avoid gastric irritation or drug inactivation. **Avoid crushing the tablet** 
4. Sublingual	Friable **small, good taste** disc for rapid **buccal disintegration**. Absorbed from buccal mucosa to systemic circulation 
5. Lozenges	**Flavored** tablet. dissolve **slowly** in mouth (for local use). 
6. Sustained release (SR) = (Spansule)	- Controled **release** or **Extended** release (ER) or Retard - Contain different coats with different disintegration time **Advantages:** 1. Rapid onset 2. long duration of action. 3. **low** frequency of administration **better** compliance **May be tablets or capsules** 
7. Effervescent Tablet	Contain sodium bicarbonate 
8. Chewable Tablet	

B. CAPSULES (The drug enclosed in a gelatin container to mask bad taste)

1. Ordinary	***Hard*** gelatin (Powder) or ***Soft*** gelatin (Liquid)
2. Spansule	sustained release capsule.
3. Enteric coated capsule:	disintegrated in the intestine to avoid gastric irritation

C. POWDER

Powder dispensed in paper packets and may be formulated as **Effervescent powder** → release CO2 → **mask the salty or bitter taste of the drug**. E.g. Magnesium sulfate 

RECTAL DOSAGE FORM

A. SOLIDS (SUPPOSITORY): DRUG INCORPORATED IN WAX

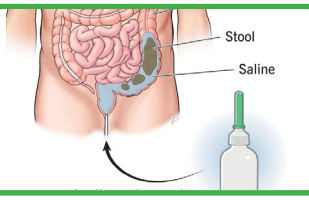
1. Local	**Glycerin:** for evacuation of lower GIT in case of constipation
2. Systemic	**Paracetamol:** Medicated suppositories

2. LIQUID (ENEMA):

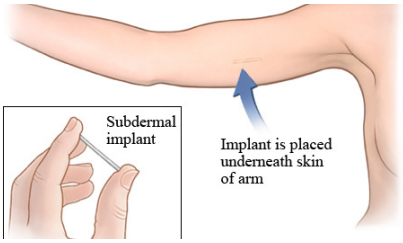
Evacuation enema	Retention enema
• Warm H_2O • 600 ml • High Pressure • **Aim:** Rectal evacuation	• Warm H_2O + the drug • 100 ml • Low Pressure • **Aim:** drug administration.

Precautions before retention enema

1. First cleaned by evacuant enema
2. Must not exceed 100 ml with low pressure
3. Drug should not be irritant



II. PARENTERAL DOSAGE FORMS

1. Liquid	**- Ampoules** (for single dose) **- vials** (multiple doses- for Unstable drugs needed solvents) 
2. Solid (Implants):	**Pellets:** Solid, Injection dosage form: e.g. Contraceptive introduced under the skin (SC) to release the drug to circulation for long time. 
3. Pre-filled Syringe:	For drug requiring accurate dosing: e.g. Enoxaprin (anticoagulants). 

III. INHALATION DOSAGE FORMS

1. GAS	**0_2 or Inhalation anesthesia**
2. VAPOR	**Highly volatile liquids** (e.g. amyl nitrite).
3. Powder	Taken through Dry power inhaler e.g. (Spinhaler)
4. Aerosols:	Suspension of a liquid or solid in a gas. **Administered through:**

Nebulizer	Metered Dose Inhaler (MDI)
- A system to deliver the drug into the lung in a form of aerosol through **jet of air.** - **Dose limitation:** by toxicity of the drug only (safer method). 	- A system to deliver the liquid drug in a form of aerosol by **propellant** added to the drug. **Disadvantages:** - Need cooperative patient - **Dose limitation** by: (less safe) • Toxicity of the drug. • Toxicity of additives (Propellant). 

IV. EXTERNAL DOSAGE FORMS

A. TOPICAL SKIN FORMS SKIN AND MUCOUS MEMBRANES DOSAGE FORMS:	
1. Lotion:	aqueous solution for local application.
2. Cream	Emulsion of oil in water incorporates the drug: Used for semi-wet lesions.
3. Ointment	The base is (Vaseline or lanoline) incorporates the drug: Used for dry lesions
4. Solid	Powder e.g. Antifungal powder
B. SYSTEMIC SKIN FORM	
Transdermal Patch:	A patch containing the drug that released through rate controlling membrane into the skin: e.g. Nitroglycerin patch.
Intranasal	For systemic and local effect

ROUTES OF ADMINISTRATION

1. Enteral (GIT).
- •Oral
- • Buccal
- • Rectal
- • Gastric Gavage

2. Parenteral (Injection)
- • Systemic
- • Local (e.g. intracardiac)

3. InhalationRectal

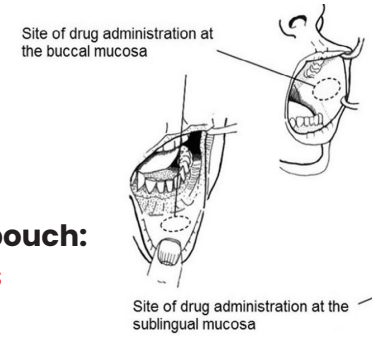
4. Topical (External).

I. ENTERAL ROUTE (GIT)

A. Buccal Route

For local or systemic effects
Lozenges (flavored tablet slowly disintegrated in in mouth for local effect)

Advantages	Examples:
1. **Easy** administration 2. **No First pass** metabolism → High bioavailability (F). 3. **Rapid** onset → suitable in emergency. 4. Can be easily **removed** in case of overdose	- **Sublingual:** Nitroglycerin (Systemic) - **In the buccal pouch:** Lozenges (Local) - **Transbuccal patch:** Fentanyl (Systemic)



Transbuccal patch:
Fentanyl

In the buccal pouch:
Lozenges

Sublingual:
Nitroglycerin

B. Oral Route

Advantages	Examples:
1. Safer 2. Self medication 3. Sterilization (No need) 4. Compliance (Good)	1. **Delayed** onset of action due to gastric emptying time 2. **FPM** → Variable bioavailability 3. **Not suitable** for **all patients** = # of oral route 4. **Not suitable** for **some drugs**. e.g. • Highly **Irritant** drugs. • Drugs **destroyed** in GIT: e.g. catecholamines. • Highly **hydrophilic** drugs: e.g. Gentamycin(not absorbed)

Contraindications:
- Emergency
- Vomiting
- Convulsion and coma

C. Rectal Route

Advantages	Disadvantages:
1. Less **FPM** than oral (Oral > rectal > SL) 2. Suitable to: - Drugs destroyed in stomach or intestine - Irritant drugs for the stomach - Both children and adult. - Patients: in emergency, in coma, with convulsion or vomiting. = # of oral	1. **Irregular** bioavailability. 2. **Inacceptable** = disagreeable (Non compliance). 3. Inflamation (**Proctitis**). With chronic use of irritant drugs

Examples: NSAIDs, Paracetamol,....

D. Gastric Gavage	E. Gastric lavage
Introduction of drugs or food into stomach by flexible tube. - Used in coma or **difficulty in swallowing**	**Washing** out stomach. - Used in **poisoning.**

II. PARENTERAL (INJECTION)

I . Intravenous route

Site of injection:
Adult: Superficial vein in the dorsum of the hand and forea...
For long term administration: Subclavian common internal or external jugular vein

Advantages	Disadvantages: "FAT IAV"
- **Immediate** onset → Suitable for emergency. - 100% **bioavailability** (no FPM) - Suitable for **Large volume** (infusion), irritant or drugs destroyed in stomach. - **Suitable** for most patients: in coma, vomiing,...	- Wrong medication (oily) → **Fat embolism**. - Wrong technique → **Air embolism** - Prolonged infusion → risk of **Thrombophlbitis** - Introduction of infection → **Bacteremia** - **Anaphylaxis** e.g. Penicillin. - **Extra-vasation** → Local Necrosis with vasconstrictors

Volume: less than 1ml to Liters

2 . Intramuscular Route

Site of injection:
Dorso-gluteal, vastus lateralis, rectus fuerous

Advantages	Disadvantages:
- **Rapid** absorption - Higher **Bioavailability** (IV > IM> oral) - Suitable for **depot** action - Suitable for **oily** preparation. (to prolong the action of the drug)	- Not suitable for **large volume** (>5 ml) and for **irritant** drugs (e.g. Glucose in hypoglycemia) - **Accidental** IV injection. (dangerous if oily). - Formation of **absces**. (sterile or infected)

Volume < 5ml to Liters

3 . Subcutaneous Route

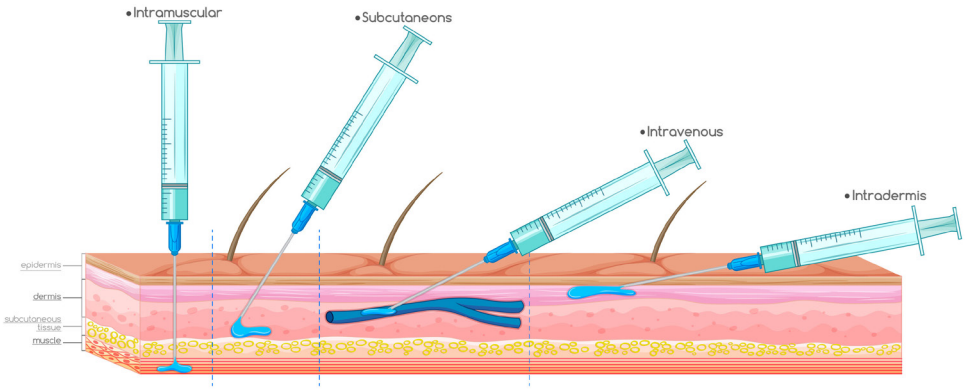
Site of injection:
Internal aspect of upper arm
Thigh& abdomen (for frequent administration)

Advantages	Disadvantages:
- Prolonged duration of action. - Acceptable for self administration (e.g. Insulin)	- Lipo-Dystrophy at injection site (atrophy or hypertrph Hypertrophy due to repeated inj. in the same site). - Irritant. - Poor absorption in peripheral circulation

Volume: < 2 ml

4 . Intradermal Route

Site of injection:
Ventral forearm in the site of choice
Injection between epidermis and dermis.

Indications	Injection Techniques:
1. Allergic tests 2. Vaccinations.	

III . INHALATION ROUTE

Can be used for local and systemic effects

Advantages	Disadvantages:
- High absorption (Large surface area). - Immediate onset of action. - High Concentration in respiratory tract. - low incidence of systemic side effects.	- Need cooperative patient (MDI). - Irregular absorption - Respiratory irritation - Cortisone→ candidiasis & Hoarseness of voice 

Volume: less than 1ml to Liters

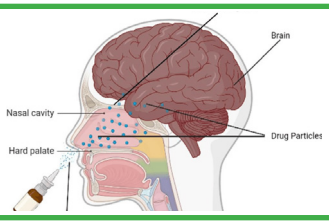
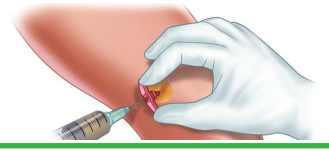
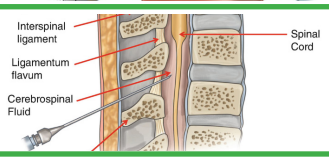
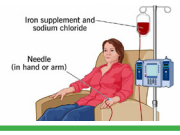
IV. TOPICAL ROUTE (SKIN OR MUCOUS MEMBRANE)

Ordinary topical application for local effect e.g Cream, ointment
Transdermal patch → for systemic effect

Examples

1. Nitroglycerin for angina
2. Hyoscine for motion sickness

Advantages	Disadvantages:
- **Gradual** onset and long duration of action. - **No** First pass Metabolism (FPM). - **No** GIT irritation. - **No** Destruction by gastric enzymes.	- Local allergy. - Variable absorption. - Plaster may drop off.

Intranasal	- Systemic effect - local effect	
Intra articular	Injection into synovial fluid of the joint for local effect	
Intra thecal	Injection into subarachnoid space contain cerebrospinal fluids	
Intravenous Infusion	Large volume Fluids or Blood Dilution of drugs	

1- Choose the correct answer about this route of administration:



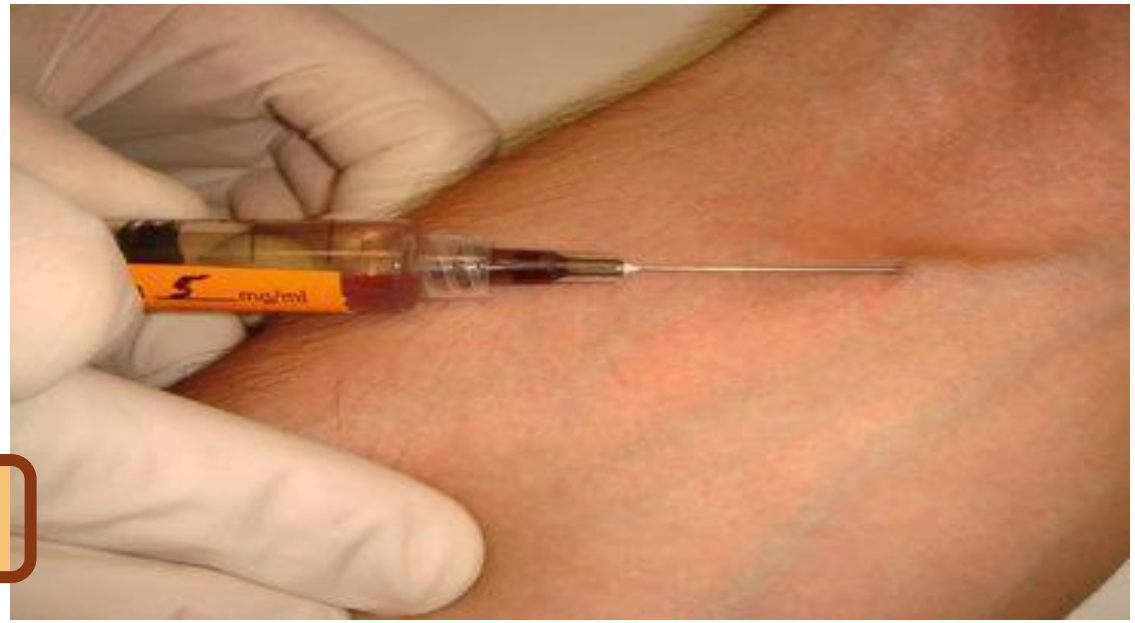
A. Has 1st pass metabolism
B. It needs well trained person
C. It has irregular bioavailability
D. Can't be used in large volumes

In emergency, the best roue to use is ……

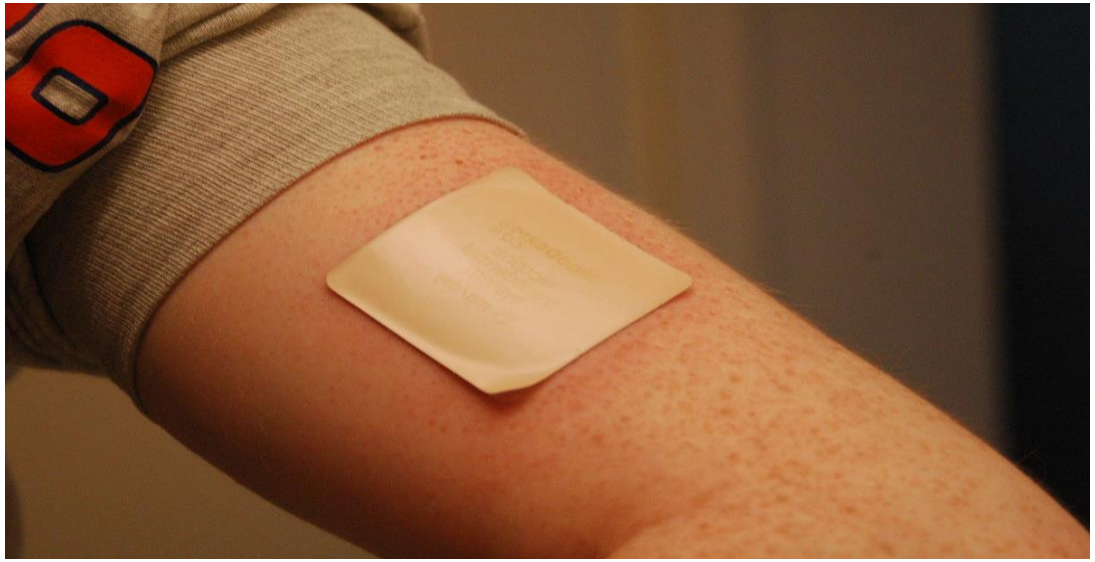
A



B



C

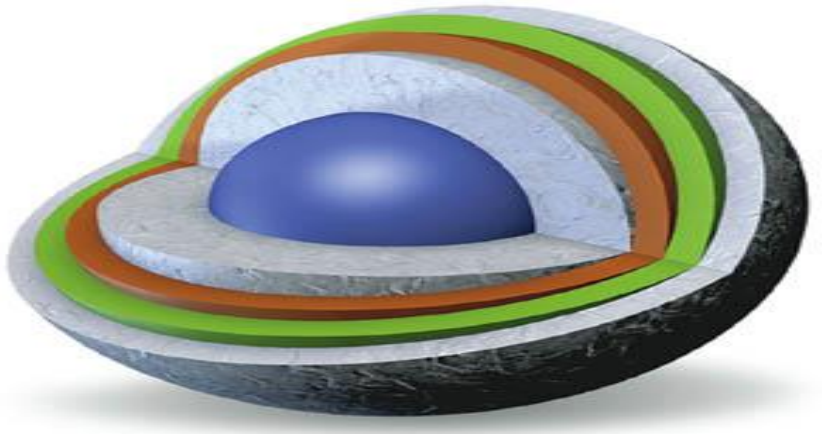


D



Regarding this dosage form, the false statement is





A. It is an oral route of administration

B. Has long duration of action

C. Characterized by low frequency of administration, and high patient compliance

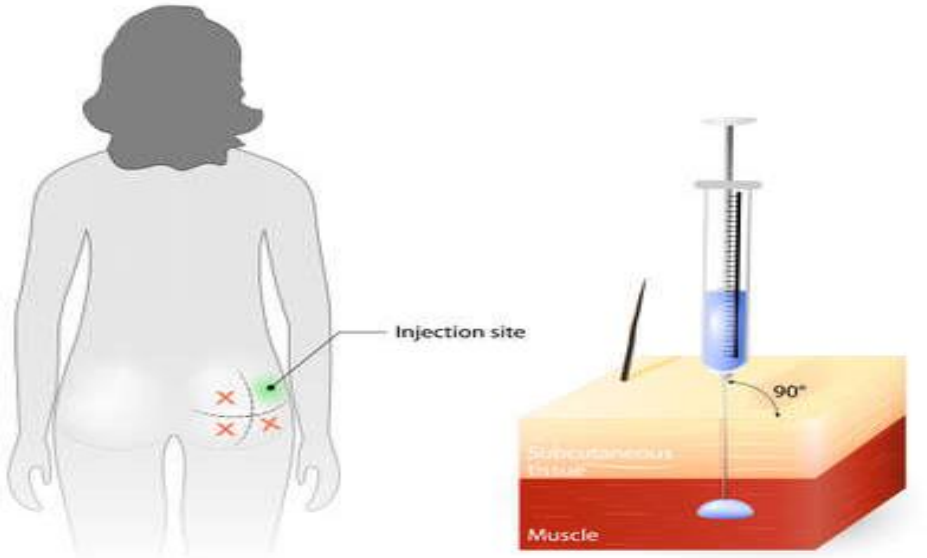
D. Can be crushed before use

Large volumes (500 ml) can be given by which of the following routes?

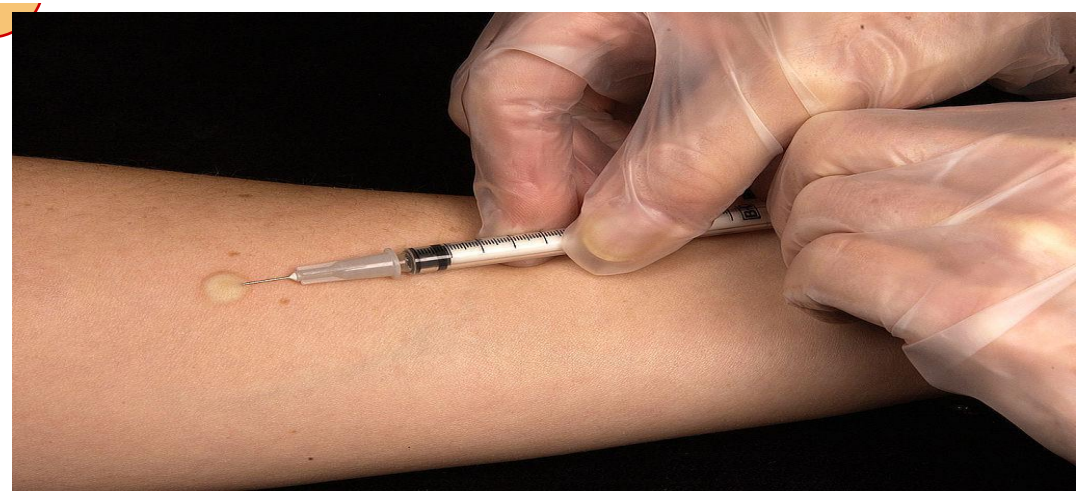
A



B



C



D



Identify the following dosage form:



A. Soft Gelatin Capsule
B. Spansule
C. Enteric-coated tablet
D. Effervescent tablet
E. Pill

Which one of these routes characterized by extensive First Pass Metabolism in some drugs ?

A



B



C



D



Which one of the following routes not suitable in presence of coma?

A



C



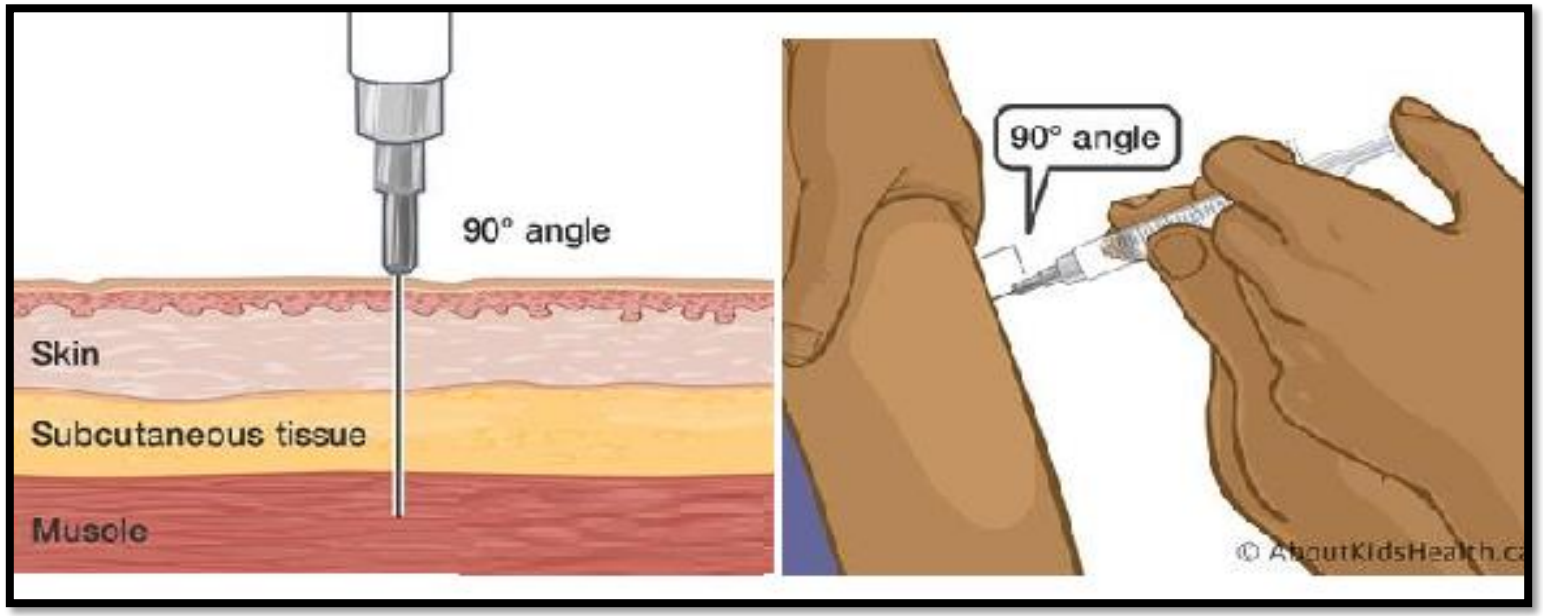
B



D

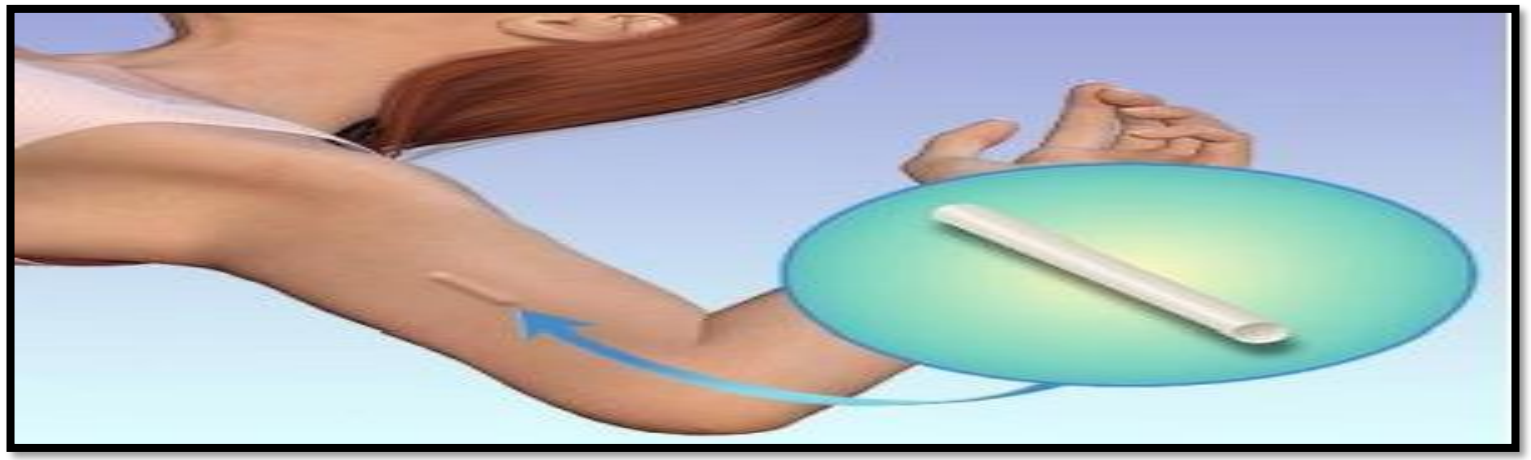


6- Choose the correct answer about this route of administration:



A. It is not painful
B. It is suitable for depot preparations
C. It produces an immediate response
D. It is the most suitable route for emergencies
E. It needs strict sterile conditions

7-For this route of administration, Choose the correct answer:



A) It is one of the enteral routes

B) It is administered intramuscular

C) It is placed under the skin

D) It is suitable for emergencies